# Mastering R Basics

## Key Terms, Core Functions & Essential Packages

R Training Workshop

Data Science & Analytics

September 17, 2026

# Overview & Core Foundations

## Training Foundation

Mastering R requires a solid foundation in core data structures, key data manipulation verbs, basic analytical functions, and the grammar of graphics.

- **1. Data Structures**: Vectors, Data Frames, Factors, Lists
- **2. Manipulation**: Filtering, transforming, aggregating
- **3. Analysis**: Summary statistics & basic testing
- **4. Visualization**: Layered graphics with `ggplot2`

# Core R Concepts & Data Structures

## Vector

Fundamental 1D container holding elements of the same data type.

- **Functions**: `c()`, `seq()`, `rep()`, `length()`, `class()`

## Data Frame

2D tabular structure where columns hold different data types.

- **Functions**: `data.frame()`, `str()`, `dim()`, `head()`, `tail()`, `names()`

## Factor

Vector type used for categorical data with defined levels.

- **Functions**: `factor()`, `levels()`, `as.factor()`

## List

Flexible object holding mixed types, vectors, or nested lists.

- **Functions**: `list()`, `str()`

# Data Manipulation & Wrangling (`dplyr` & `tidyr`)

- **Pipe Operator (|> or %>%)**: Passes the output of one function directly into the next for readable workflows.
- **Selection & Filtering**: Subsetting variables or observations based on specific criteria.
  - **Key Functions**: `select()` (columns), `filter()` (rows)
- **Transformation**: Modifying existing columns or creating new calculated fields.
  - **Key Functions**: `mutate()`, `rename()`, `arrange()`
- **Aggregation**: Grouping data by categorical variables to compute group-level summaries.
  - **Key Functions**: `group_by()`, `summarize()`, `count()`, `n()`
- **Reshaping**: Converting datasets between wide and long formats.
  - **Key Functions**: `pivot_longer()`, `pivot_wider()`

# Data Analysis & Summary Statistics

## Descriptive Statistics

Calculating central tendency, dispersion, and overall distribution.

- **Key Functions**: `mean()`, `median()`, `sd()`, `quantile()`, `summary()`

## Frequency & Cross-Tabulation

Counting categorical counts and relative proportions.

- **Key Functions**: `table()`, `prop.table()`

## Basic Statistical Tests

Examining correlations and basic linear relationships.

- **Key Functions**: `cor()`, `t.test()`, `lm()`

# Data Visualization (ggplot2)

- **Aesthetics (aes)**: Mapping dataset variables to visual dimensions like position $(x, y)$, color, fill, and size.
- **Geometries (geom_*)**: Declaring the geometric objects used to render data points:
  - `geom_point()`: Scatter plots
  - `geom_bar()`: Frequency counts
  - `geom_histogram()`: Continuous distributions
  - `geom_boxplot()`: Outliers and IQR distributions
  - `geom_line()`: Trend lines
- **Faceting**: Splitting graphics into multiple subplots by category.
  - **Key Functions**: `facet_wrap()`, `facet_grid()`

# Essential Package Ecosystem

| Package | Primary Role |
|---|---|
| `tidyverse` | Master library bundling `dplyr`, `ggplot2`, `tidyr`, `readr`, and `stringr` |
| `readr` / `readxl` | Reading CSV and Excel spreadsheets into clean data frames |
| `dplyr` | Core grammar for fast data manipulation |
| `tidyr` | Functions for data cleaning and reshaping |
| `ggplot2` | Layered visualization framework |